AWS Networking — Route Table

1. What is a Route Table?
A Route Table contains rules that decide where network traffic should go.
It tells AWS where to send traffic from a subnet.
Simple ga gurthupettuko:
Route Table ante network traffic ekkadiki vellalo decide chese rules.
Eng: A Route Table contains rules that decide where network traffic should go.

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2. What is a Route?
A Route is a rule inside a Route Table that defines the destination of network traffic.
It tells AWS where the traffic should be forwarded.
Simple ga gurthupettuko:
Route ante traffic ekkadiki vellalo cheppe rule.
Eng: A Route defines where network traffic should be forwarded.

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3. What is a Default Route?
A Default Route is used when there is no specific route available for the destination.
0.0.0.0/0 is commonly used as the default route for IPv4 traffic.
Simple ga gurthupettuko:
Default Route ante specific route lekapothe use ayye route.
Eng: A Default Route is used when no specific route is available.

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4. What is Route Table Association?
Route Table Association connects a Route Table to a subnet.
The subnet then uses that Route Table for its network traffic.
Simple ga gurthupettuko:
Association ante Route Table ni Subnet ki connect cheyyadam.
Eng: Route Table Association connects a Route Table to a subnet.

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5. Can one Route Table be associated with multiple Subnets?
Yes, one Route Table can be associated with multiple subnets.
All those subnets can use the same routing rules.
Simple ga gurthupettuko:
Oka Route Table ni multiple Subnets ki connect cheyyachu.
Eng: One Route Table can be associated with multiple subnets.

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6. What is a Main Route Table?
The Main Route Table is the default Route Table of a VPC.
A subnet uses it when no other Route Table is explicitly associated with it.
Simple ga gurthupettuko:
Main Route Table ante VPC ki default routing table.
Eng: The Main Route Table is the default Route Table of a VPC.

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7. What is a Custom Route Table?
A Custom Route Table is a Route Table that we create for specific routing requirements.
We can associate it with one or more subnets.
Simple ga gurthupettuko:
Custom Route Table ante mana requirement kosam create chesukune routing table.
Eng: A Custom Route Table is created for specific routing requirements.

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8. How do you make a Subnet public using a Route Table?
We add a route to the Internet Gateway in the Route Table associated with the subnet.
The resources also need the required public IP configuration to access the internet.
Simple ga gurthupettuko:
Public Subnet kosam Route Table lo Internet Gateway route undali.
Eng: A public subnet needs a Route Table route to an Internet Gateway.

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9. How does a Private Subnet access the Internet?
A private subnet can use a NAT Gateway for outbound internet access.
The Route Table sends internet traffic from the private subnet to the NAT Gateway.

Simple ga gurthupettuko:
Private Subnet Internet kosam NAT Gateway ni use chestundi.
Eng: A private subnet can use a NAT Gateway for outbound internet access.

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10. What is 0.0.0.0/0**?**
0.0.0.0/0 represents all IPv4 addresses.
It is commonly used as a default route for internet-bound traffic.
Simple ga gurthupettuko:
0.0.0.0/0 ante all IPv4 traffic.
Eng: 0.0.0.0/0 represents all IPv4 addresses.

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11. What happens if a Subnet has no explicit Route Table?
If a subnet has no explicit Route Table association, it uses the VPC's Main Route Table.
The Main Route Table therefore provides the default routing for that subnet.
Simple ga gurthupettuko:
Explicit Route Table lekapothe Subnet Main Route Table ni use chestundi.
Eng: A subnet without an explicit association uses the VPC's Main Route Table.

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12. How do you troubleshoot a Route Table issue?
First, check whether the subnet is associated with the correct Route Table.
Then check the required route and the target such as Internet Gateway or NAT Gateway.
Simple ga gurthupettuko:
Routing problem ante Association → Route → Gateway check cheyyali.
Eng: Check the Route Table association, route, and required gateway.

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13. What is the difference between Public and Private Route Table?
A Public Route Table normally contains a route to an Internet Gateway.
A Private Route Table normally does not have a direct route to an Internet Gateway.
Simple ga gurthupettuko:
Public Route Table = IGW route
Private Route Table = Direct IGW route ledu.
Eng: Public Route Table uses an Internet Gateway route, while Private Route Table does not use a direct Internet Gateway route.

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14. What is the most important thing to remember about Route Table?
A Route Table controls where network traffic from a subnet should go.
It works with components such as Internet Gateway, NAT Gateway, and VPC Peering.
Simple ga gurthupettuko:
Route Table ante Subnet traffic ki direction decide chestundi.
Eng: A Route Table controls where network traffic from a subnet should go.
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